

EDUCATION

Purdue University

Ph.D. in Computer Science

Advisor: Ruqi Zhang

IN, USA

Aug. 2025 – Present

Tianjin University

B.S. in Data Science and Big Data Technology

School of Mathematics · Advisors: Bing Cao and Qinghua Hu

Tianjin, China

Aug. 2021 – Jun. 2025

RESEARCH INTERESTS

I study how large language and vision-language models can improve from limited, noisy, or imperfect supervision. My work develops reliable and efficient post-training algorithms for reasoning and safety, with a focus on reinforcement learning, on-policy distillation, and self-correction.

PUBLICATIONS

* Equal contribution

Preprints and Technical Reports

[P1] Does On-Policy Distillation Really Distill? From Noisy Teacher to Self-Improvement

Yi Ding, Ruqi Zhang

Preprint, 2026 [PDF] [Project]

#1 on Hugging Face Daily Papers (Sep. 1, 2026)

[P2] Addressing Performance Saturation for LLM RL via Precise Entropy Curve Control

Bolian Li, Yifan Wang, Yi Ding, Anamika Lochab, Ananth Grama, Ruqi Zhang

Preprint, 2026 [PDF] [Project]

[P3] SafeWork-R1: Coevolving Safety and Intelligence under the AI-45° Law

Shanghai AI Lab: Yicheng Bao, ..., Yi Ding, ... (100+ authors)

Technical report, 2025 [PDF]

Conference Publications

[C1] Learning Self-Correction in Vision–Language Models via Rollout Augmentation

Yi Ding, Ziliang Qiu, Bolian Li, Ruqi Zhang

ICML 2026 · International Conference on Machine Learning [PDF] [Project]

[C2] Position: Modular Safety Guardrails Are Necessary for Foundation-Model-Enabled Robots in the Real World

Joonkyung Kim, Wenxi Chen, Davood Soleymanzadeh, Yi Ding, Xiangbo Gao, Zhengzhong Tu, Ruqi Zhang, Fan Fei, Sushant Veer, Yiwei Lyu, Minghui Zheng, Yan Gu

ICML 2026 · International Conference on Machine Learning [PDF]

[C3] Rethinking Bottlenecks in Safety Fine-Tuning of Vision Language Models

Yi Ding*, Lijun Li*, Bing Cao, Jing Shao

ICLR 2026 · International Conference on Learning Representations [PDF] [Project]

[C4] Sherlock: Self-Correcting Reasoning in Vision-Language Models

Yi Ding, Ruqi Zhang

NeurIPS 2025 · Advances in Neural Information Processing Systems [PDF] [Project]

PUBLICATIONS (CONTINUED)

[C5] **Visual Contextual Attack: Jailbreaking MLLMs with Image-Driven Context Injection**

Ziqi Miao*, Yi Ding*, Lijun Li, Jing Shao

EMNLP 2025 · *Empirical Methods in Natural Language Processing, Main Conference* [PDF]

[C6] **ETA: Evaluating Then Aligning Safety of Vision Language Models at Inference Time**

Yi Ding, Bolian Li, Ruqi Zhang

ICLR 2025 · *International Conference on Learning Representations* [PDF] [Project]

[C7] **Test-Time Dynamic Image Fusion**

Bing Cao, Yanan Xia*, Yi Ding*, Changqing Zhang, Qinghua Hu

NeurIPS 2024 · *Advances in Neural Information Processing Systems* [PDF]

[C8] **Predictive Dynamic Fusion**

Bing Cao, Yanan Xia*, Yi Ding*, Changqing Zhang, Qinghua Hu

ICML 2024 · *International Conference on Machine Learning* [PDF]

RESEARCH EXPERIENCE

RZ-Lab, Purdue University

IN, USA

Research Assistant

May 2024 – May 2025

Advisor: Ruqi Zhang

- Introduced **Sherlock**, enabling VLM self-correction with 20K annotated examples and subsequent self-improvement without external supervision.
- Proposed **ETA** to improve VLM safety at inference time by evaluating multimodal inputs and steering response generation, without additional training.

Open Trust Lab, Shanghai AI Laboratory

Beijing, China

Research Scientist

Dec. 2024 – Mar. 2025

Advisor: Lijun Li

- Created the **Multi-Image Safety (MIS)** dataset and a fine-tuning approach that teaches VLMs to reason about safety risks arising from interactions among multiple images.

MLDM Lab, Tianjin University

Tianjin, China

Research Assistant

Sep. 2023 – Dec. 2024

Advisors: Bing Cao and Qinghua Hu

- Derived generalization error bounds for dynamic multimodal fusion and developed **Predictive Dynamic Fusion** and **Test-Time Dynamic Image Fusion**.

AWARDS AND RESEARCH SUPPORT

NSF ACCESS Discover Project Award

2025–2026

Project: LLM Post-Training

ACADEMIC SERVICE

Reviewer: ICLR 2025–2027; ICML 2026; NeurIPS 2025, 2026; CIKM 2026;

ACL Rolling Review (May 2025).

SKILLS AND LANGUAGES

Technical: Python, MATLAB, $\text{\LaTeX}$; mathematical and statistical analysis.

Languages: Mandarin Chinese (native); English (TOEFL: 102).